

Statistical Compressive Sensing and Feature Extraction of Time-Frequency Spectrum From Narrowband Radar

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Aiming at the signal reconstruction problem for the conventional narrowband radar system, we propose a new statistical compressive sensing (SCS) method to achieve the reconstruction of superresolution time-frequency spectrum from the corrupted time-domain measurement. The proposed method assumes that the signal obeys complex Gaussian distribution and develops a hierarchical Bayesian model. Variational Bayesian expectation maximization (VBEM) is used to perform inference for the posterior distributions of the model parameters. In order to fully exploit the superresolution characteristics of reconstructed spectrum, a novel superresolution time-frequency feature vector is extracted for subsequent classification of ground moving targets, i.e., walking person and a moving wheeled vehicle. Experimental results on measured data show that the proposed reconstruction method can obtain good reconstruction results and the superresolution feature has good classification performance for human and vehicle targets.

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I. INTRODUCTION

Ground moving targets classification by narrowband radar is an important signal processing task, which has found widespread military and civilian applications, such as border control and monitoring system. In contrast to high-resolution radar imaging systems, a low-resolution narrowband radar can only provide Doppler information on targets. Relative motion of nonrigid parts of complex targets can induce additional features in the Doppler frequency spectrum [1], [2]. These features are called micro-Doppler effect and appear as sidebands around the central Doppler frequency. Literatures [3]–[5] show the micro-Doppler is an effective discriminative feature for the classification of moving persons. In [6], a robust method for classifying the micro-Doppler signatures of helicopters via label consistent K -singular value decomposition (K -SVD) is presented. In addition, as shown in [7] and [8], with the help of micro-Doppler signature, information on wheels and pedestrian gaits can be extracted.

Because of the time-varying properties of micro-Doppler, time-frequency analysis of nonstationary radar signals can provide information about time-dependent Doppler frequency variations and allow to extract classification features. There are lots of ways for time-frequency analysis, such as short-time Fourier transform (STFT), the Wigner distribution function, Gabor transform, of which the STFT is most widely used. The time-frequency spectrum for a target can be obtained by taking the STFT of the slow-time radar return signal. In the last decade years, several radar automatic target recognition techniques and activity classification algorithms, which are based on time-frequency spectrum, have been presented. Many features have been proposed to classify human micro-Doppler signatures [9], such as average torso velocity, Doppler bandwidth, and leg swing frequency. After analyzing the differences between the micro-Doppler signatures of pedestrians and vehicles, a novel feature extraction method is proposed to classify ground moving targets in [10]. Since higher frequency resolution can provide more details of spectrum, superresolution time-frequency feature may yield better classification performance. Chen *et al.* [11] establishes the framework of sparse time-frequency distribution and achieves high-resolution of m-D signal in time-sparse domain. Nevertheless, the existing time-frequency feature extraction methods give no consideration to the influence of frequency resolution on classification performance.

In fact, classification of ground moving targets based on physical feature generally requires a long dwell time [12]. For a narrowband surveillance radar system, which accomplishes the task of simultaneous search and classification, it is difficult to obtain the continuous measurement of a target for a long time. Usually, we can only acquire a number of discontinuous measurements, among which some signal samples are missing. Removing the requirement of long dwell times increases the possibility of simultaneous tracking and classification of multiple targets. However, most micro-Doppler signatures are the reflection of periodic

vibrations or rotations, and the periodicity, which is a significant physical feature to classify ground moving target, can hardly be observed in short dwell time. Directly using each measurement within the short dwell time will degrade classification performance of ground moving targets when using physical feature [13], [14].

Therefore, in this paper, we study on reconstructing the time-frequency feature of an original complete signal from the existing discontinuous measurement signal, which is referred to as the corrupted signal in this paper, for the subsequent classification. Compressive sensing (CS) [15]–[19] is a new theory, which breaks the limitation of the conventional Nyquist–Shannon sampling theorem and can reconstruct the complete signal from few data by exploiting the sparsity as the prior knowledge of the original signal. It is well known that if an n -dimensional signal admits an s -sparse representation, then one can reconstruct exactly the original signal with $m = O(s \log n/s)$ measurements via l_1 minimization method, i.e., the basis pursuit (BP) or iterative methods such as matching pursuit (MP) and orthogonal matching pursuit algorithm (OMP) [20], [21]. In [22], the time-frequency spectrum is reconstructed by using multiwindow spectrogram approach in conjunction with sparse signal recovery techniques that exploit the multiple measurement vector model. In [23], the algorithm for CS reconstruction of signals with sinusoidal phase modulation is proposed. With respect to signal sparsity assumption on the dictionary, typically an orthogonal basis, redundant and signal adaptive dictionaries further improve the CS performance. In [24], a novel reconstruction algorithm that uses K -SVD dictionary learning is proposed to reconstruct magnetic resonance images from undersampled k -space data.

In contrast to the traditional CS methods that only utilize the underlying sparse property of signal, statistical compressed sensing (SCS) [25] also exploits the statistical information of signal. In [25], the complete training signals have to be used to estimate the Gaussian mixture model (GMM) and the quality of the estimation depends heavily on the training samples. However, acquiring ideal training samples is a great challenge for us in a realistic radar scenario. This problem is addressed by directly learning the signal model from the measurements, i.e., the corrupted signals, in [26].

It is worth noting that the proposed algorithm in [26] is applied to the real-valued image data or video data. However, the target echo data are generally complex-valued in radar signal processing. Besides, in the existing CS methods, both the measurement signal and the reconstructed signal are in the same feature domain. For the subsequent classification, the feature transformation usually needs to be done again to extract feature from the reconstructed signal.

In this paper, we propose a complex-valued signal reconstruction algorithm that directly reconstructs the original signal's superresolution time-frequency spectrum. The proposed method assumes that the complete signal obeys complex Gaussian distribution. Then, a hierarchical Bayesian model is developed to learn complex Gaussian distribution via the corrupted time-domain measurement,

where no complete training samples are required. The reconstruction of superresolution time-frequency spectrum is achieved by calculating the posterior distributions of the model parameters. Efficient inference is performed via variational Bayesian expectation maximization (VBEM). In addition, a novel two-dimensional micro-Doppler feature vector extracted from the superresolution time-frequency spectrums is defined in this paper to classify the ground moving targets. The proposed feature takes advantage of the dispersive degree of the main energy component and has a higher degree of discriminative in the superresolution time-frequency spectrum. Experimental results show that the proposed reconstruction method can obtain good reconstruction results and the superresolution feature has good classification performance.

The organization of this paper is as follows. In Section II, we propose a new signal reconstruction model under the frame of SCS and the superresolution time-frequency spectrums are obtained. In Section III, a novel feature extraction method for superresolution time-frequency spectrums is proposed. In Section IV, the reconstruction performance of our method and conventional CS algorithm are compared. Besides, support vector machine (SVM) classifier is exploited to evaluate the classification performance of our proposed features. The conclusion is followed in Section V.

II. STATISTICAL SIGNAL RECONSTRUCTION FOR COMPLEX GAUSSIAN SIGNAL

A. Compressive Sensing

Compressible sensing aims at achieving accurate signal reconstruction while sampling signals at a low sampling rate, typically far smaller than that of Nyquist. The general problem of signal reconstruction can be expressed as

$$\mathbf{y} = \Phi \mathbf{x} + \boldsymbol{\varepsilon} \quad (1)$$

where $\mathbf{y} \in \mathbb{R}^{K \times 1}$ is the corrupted measurements, $\mathbf{x} \in \mathbb{R}^{N \times 1}$ is a signal of interest, $\Phi \in \mathbb{R}^{K \times N}$ ($K \ll N$) is a measurement matrix, and $\boldsymbol{\varepsilon}$ is an additive noise. In order to ensure recovery from the measurements, it is generally assumed that $\mathbf{x}$ is compressible or sparse, i.e., we can express the signal $\mathbf{x}$ as

$$\mathbf{x} = \sum_{i=1}^M \Psi_i \mathbf{a}_i = \Psi \mathbf{a} \quad (2)$$

where $\Psi = [\Psi_1, \Psi_2, \dots, \Psi_M] \in \mathbb{R}^{N \times M}$ ($N < M$) is a sparse basis or dictionary, $\mathbf{a} = [a_1, a_2, \dots, a_M] \in \mathbb{R}^{M \times 1}$ is a sparse coefficient vector. Equation (2) has just a few large coefficients and many small coefficients.

If Φ is chosen properly and $\mathbf{x}$ is sparse, $\mathbf{x}$ can be reconstructed from $\mathbf{y}$ by solving the optimal problem as follows:

$$\hat{\mathbf{a}} = \arg \min \|\mathbf{a}\|_0 \text{ s.t. } \mathbf{y} = \Phi \Psi \mathbf{a} = \hat{\Phi} \mathbf{a} \quad (3)$$

where $\hat{\Phi} = \Phi \Psi$ is a sensing matrix satisfies restricted isometry property. Then, we can obtain the reconstruction of

signal $\mathbf{x}$

$$\hat{\mathbf{x}} = \Psi \hat{\mathbf{a}}. \quad (4)$$

Three widely used reconstruction algorithms for solving the (3) are the BP, the MP, and the OMP algorithm.

Actually, the performance of CS algorithms crucially relies on the sparsity of the signals. Hence, a more compact representation for the signals leads to more improvement in performance. How to choose the efficient dictionary is a decisive question in the sparse-representation problem and dictionary learning is the feasible method to adapt the dictionary to the data. Dictionary learning algorithms iteratively perform the two stages that include sparse coding and dictionary update. In the first stage, the sparse approximation of the signals is computed by using the current dictionary. The following stage is the update of the dictionary. The most popular one used to learn the dictionary is K -SVD algorithm [27].

B. Statistical CS

SCS is a novel framework of CS, which assumes that the complete signal obeys a probability distribution and can achieve accurate reconstruction. The mean square error (MSE) is exploited to describe the reconstruction error between $\hat{\mathbf{x}}$ and $\mathbf{x}$ as

$$J = E_{\mathbf{x}|\mathbf{y}} (\|\mathbf{x} - \hat{\mathbf{x}}\|_2^2) = \int \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 p(\mathbf{x}|\mathbf{y}) d\mathbf{x} \quad (5)$$

where $E_{\mathbf{x}|\mathbf{y}}(\cdot)$ denotes the expectation of variable with respect to $\mathbf{x}$ under the condition that $\mathbf{y}$ is given, and $p(\mathbf{x}|\mathbf{y})$ is the posterior distribution of $\mathbf{x}$ under the condition that $\mathbf{y}$ is given. Here, $p(\mathbf{x}|\mathbf{y}) = p(\mathbf{x})p(\mathbf{y}|\mathbf{x})/\int p(\mathbf{x})p(\mathbf{y}|\mathbf{x})d\mathbf{x}$ with $p(\mathbf{x})$ denoting the prior distribution of $\mathbf{x}$ and $p(\mathbf{y}|\mathbf{x})$ denoting conditional distribution of $\mathbf{y}$. The optimal reconstructed signal, which is obtained by taking the derivative of (5), is based on a minimum mean square error criterion, can be expressed as

$$\hat{\mathbf{x}} = E_{\mathbf{x}|\mathbf{y}}(\mathbf{x}). \quad (6)$$

The heart of such reconstruction is how to obtain the prior distribution of unknown complete signal. The theoretical results under the SCS framework in [25] and [26] are based on the assumption that the complete signal is drawn from the GMM.

C. Signal Reconstruction Model

1) *Model Description:* The general radar signal reconstruction model based on the SCS framework can be expressed as

$$\mathbf{y} = \Phi \mathbf{X} + \boldsymbol{\varepsilon} \quad (7)$$

where $\mathbf{X} \in \mathbb{C}^{D \times 1}$ is the complete radar time-domain signal, $\mathbf{y} \in \mathbb{C}^{D \times 1}$ is the corrupted time-domain signal, $\boldsymbol{\varepsilon} \in \mathbb{C}^{D \times 1}$ is the additive noise, $\Phi \in \mathbb{R}^{D \times D}$ is a measurement matrix. Here, Φ is a diagonal matrix with the diagonal element being valued from $\{0, 1\}$, with the probability of drawing 0 defined by the rate of missing.

As a linear transform, the STFT of the signal $\mathbf{X}$ can be modeled in a vector form $\mathbf{s}$, shown as

$$\mathbf{s} = \mathbf{T} \mathbf{X} \quad (8)$$

where $\mathbf{T}$ is the STFT matrix. The signal $\mathbf{X}$ can be expressed as

$$\mathbf{X} = \mathbf{G} \mathbf{s} \quad (9)$$

where $\mathbf{G}$ is the pseudoinverse of $\mathbf{T}$, i.e., the inverse STFT matrix. Note that matrix $\mathbf{G}$ can be expressed as $\mathbf{G} = \mathbf{W} \mathbf{F}$ [28]. $\mathbf{F}$ is a block diagonal matrix with the form

$$\mathbf{F} = \begin{bmatrix} \mathbf{F}_n & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \mathbf{F}_n \end{bmatrix} \quad (10)$$

where $\mathbf{F}_n$ is the inverse Fourier transform matrix $[\mathbf{F}_n]_{k,l} = (1/L) \exp(j2\pi kl/L)$, $k = 0, 1, \dots, M-1$, $l = 0, 1, \dots, L-1$, M is the length of time window, L is the length of the Fourier transform, $\mathbf{W} \in \mathbb{R}^{D \times (M \times (D-M+1))}$ is a sparse matrix which restores the signal. It is worth noting that $\mathbf{s}$ is the vectorization of the time-frequency spectrum and is sparse.

In order to eliminate the step for making time-frequency analysis on reconstructed signal and reconstruct the time-frequency spectrum directly, an inverse STFT matrix $\mathbf{G}$ is introduced in the present reconstruction model. The reconstructed time-frequency spectrum can be directly used to carry out the subsequent ground moving target classification. Equation (8) can be transformed into

$$\mathbf{y} = \Phi \mathbf{G} \mathbf{s} + \boldsymbol{\varepsilon} = \Psi \mathbf{s} + \boldsymbol{\varepsilon} \quad (11)$$

where $\Psi = \Phi \mathbf{G}$, $\mathbf{G} \in \mathbb{C}^{D \times ((D-M+1) \times L)}$, and $\mathbf{s} \in \mathbb{C}^{((D-M+1) \times L) \times 1}$.

The radar target echo acquired by narrowband radar can be regarded as the superposition of a large number of mutually independent scattered point echoes. On the basis of the central limit theorem, the in-phase component and quadrature component of the radar echo obey an approximate Gaussian distribution, i.e., the corresponding complex random variables obey the complex Gaussian distribution. Therefore, the signal $\mathbf{X}$ is assumed to follow a complex Gaussian distribution. As shown in Fig. 1, the histograms of the in-phase and quadrature components of the complex echoes from the measured data can approximate to Gaussian distributions, which means this assumption can work well for the measured complex data.

Because $\mathbf{s}$ is a linear transformation of $\mathbf{X}$, $\mathbf{s}$ can also be assumed to follow a complex Gaussian distribution $\mathcal{CN}(\mathbf{u}, \mathbf{D}^{-1})$, where $\mathbf{u}$ and $\mathbf{D}^{-1}$ are, respectively, mean and covariance. The probability density function (PDF) of $\mathbf{s}$ has a complex Gaussian form as

$$p(\mathbf{s}|\mathbf{u}, \mathbf{D}^{-1}) = |\mathbf{D}| \exp[-(\mathbf{s} - \mathbf{u})^H \mathbf{D}(\mathbf{s} - \mathbf{u})] / \pi^D. \quad (12)$$

For the noise component, it is independent of the signal component. Here, the noise component $\boldsymbol{\varepsilon}$ is assumed to be the complex Gaussian white noise, which follows a zero mean complex Gaussian distribution $\mathcal{CN}(\mathbf{0}, \gamma^{-1} \mathbf{I})$

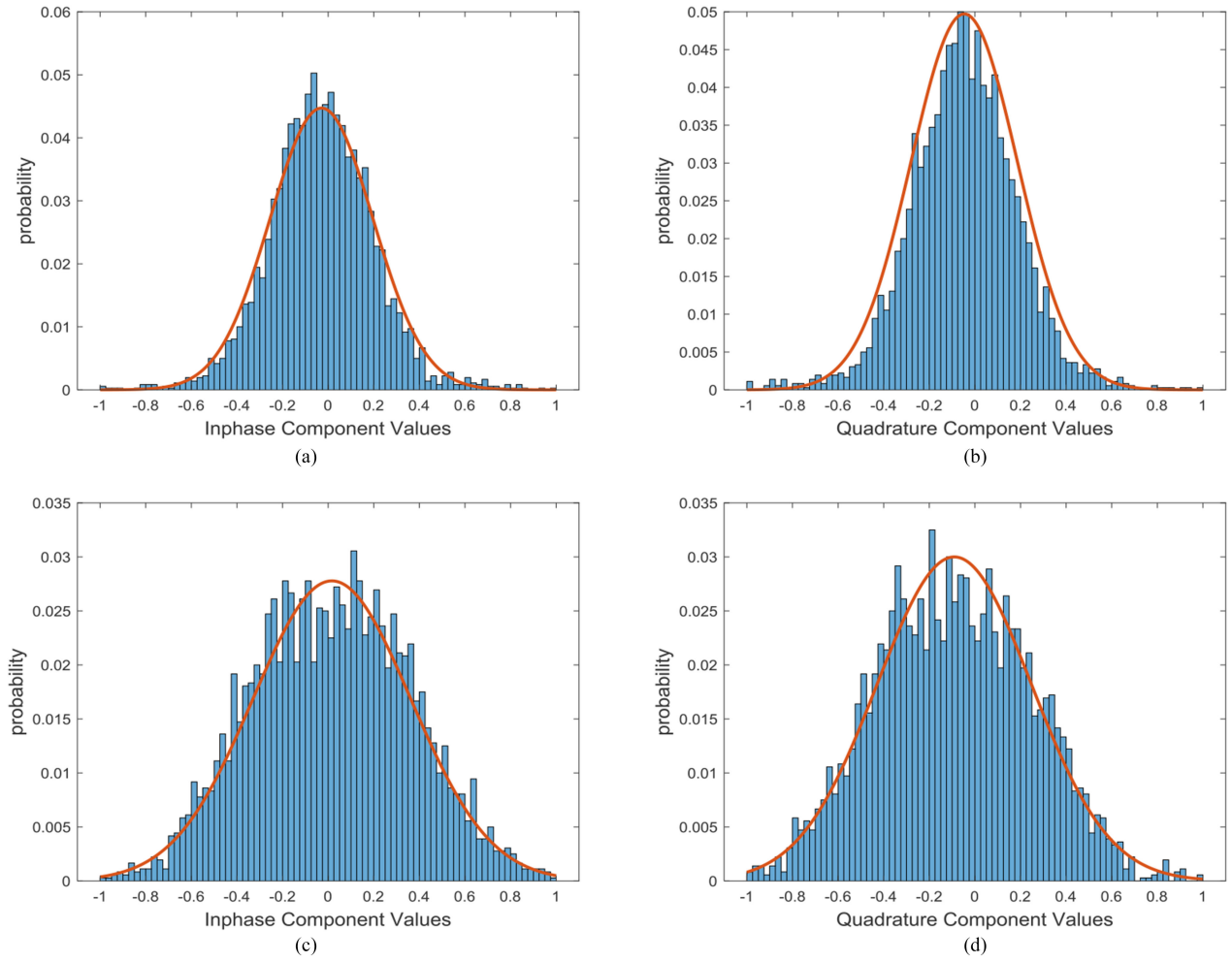


Fig. 1. Distribution examples of the inphase and quadrature components of the complex echoes. The bars represent the histograms, and the red lines represent the Gaussian PDF with the mean and variance determined by the measured data. (a) In-phase components of a walking person. (b) Quadrature components of a walking person. (c) In-phase components of a moving vehicle. (d) Quadrature components of a moving vehicle.

with unknown noise power γ following a gamma distribution $\text{Gamma}(\gamma|a_0, b_0)$. The PDF of γ can be expressed as

$$p(\gamma) \sim \text{Gamma}(\gamma|a_0, b_0) = \Gamma(a_0)^{-1} b_0^{a_0} \gamma^{a_0-1} e^{-b_0\gamma} \quad (13)$$

where $\Gamma(a_0) = \int_0^\infty t^{a_0-1} e^{-t} dt$.

In order to build up the complete Bayesian framework, we define a conjugate prior distribution for parameters $\mathbf{u}$ and $\mathbf{D}$: $\mathbf{u} \sim \mathcal{CN}(\mathbf{m}_0, (\beta_0 \mathbf{D})^{-1})$, $\mathbf{D} \sim W(v_0, \mathbf{W}_0)$, where $W(\cdot)$ denotes wishart distribution. The PDF of $\mathbf{u}$ and $\mathbf{D}$ can be expressed as

$$p(\mathbf{u} | \mathbf{m}_0, (\beta_0 \mathbf{D})^{-1}) = \frac{\exp(-(\mathbf{u} - \mathbf{m}_0)^H \beta_0 \mathbf{D} (\mathbf{u} - \mathbf{m}_0))}{(\pi^D |(\beta_0 \mathbf{D})^{-1}|)} \quad (14)$$

$$p(\mathbf{D} | v_0, \mathbf{W}_0^{-1}) = \frac{(|\mathbf{W}_0|^{-v_0/2} |\mathbf{D}|^{(v_0-D-1)/2} \exp(-tr(\mathbf{W}_0^{-1} \mathbf{D})/2))}{Z} \quad (15)$$

$$Z = 2^{v_0 D/2} \pi^{D(D-1)/4} \prod_{k=1}^D \Gamma((v_0 - k + 1)/2)$$

where $tr(\cdot)$ denotes the trace of matrix.

The signal model is summarized as follows:

$$\begin{aligned} \mathbf{y} &= \Psi \mathbf{s} + \boldsymbol{\varepsilon} \quad \boldsymbol{\varepsilon} \sim \mathcal{CN}(\mathbf{0}, \gamma^{-1} \mathbf{I}) \\ \mathbf{s} &\sim \mathcal{CN}(\mathbf{u}, \mathbf{D}^{-1}) \quad \mathbf{u} \sim \mathcal{CN}(\mathbf{m}_0, (\beta_0 \mathbf{D})^{-1}) \\ \mathbf{D} &\sim W(v_0, \mathbf{W}_0) \quad \gamma \sim \text{Gamma}(a_0, b_0). \end{aligned} \quad (16)$$

A graphical representation of the model we proposed is illustrated in Fig. 2.

2) *Model Inference*: In this section, we derive a VBEM algorithm [29] to perform inference for our reconstruction model.

We can get a set of truncated signal $\{\mathbf{y}_i\}_{i=1}^N$ by a sliding window on the corrupted time-domain radar signal. The time-frequency spectrums $\{\mathbf{s}_i\}_{i=1}^N$ can be reconstructed by using linear measurements $\{\mathbf{y}_i\}_{i=1}^N$. The superresolution reconstruction of time-frequency spectrum means that the value of L is larger than that of M .

The variables are divided into three categories: corrupted signal $\boldsymbol{\Gamma} = \{\mathbf{y}_i\}_{i=1}^N$, implicit variable $\boldsymbol{\Theta} = \{\mathbf{s}, \mathbf{u}, \mathbf{D}, \gamma\}$, parameter $\boldsymbol{\Omega} = \{a_0, b_0, v_0, \mathbf{W}_0\}$.

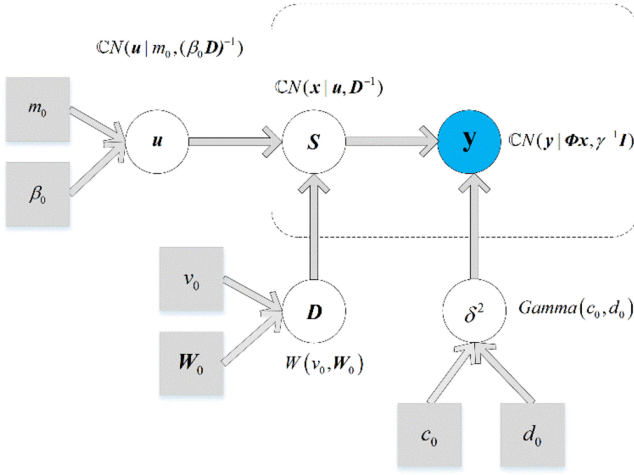


Fig. 2. Directed acyclic graph of the model.

Log-likelihood function of the model can be expressed as

$$\begin{aligned}
 Q(\Theta, \Omega) = & LN \ln \gamma \\
 & - \gamma \sum_{i=1}^N \left(\frac{y_i^H y_i - y_i^H \Psi_i s_i - s_i^H \Psi_i^H y_i}{+ s_i^H \Psi_i^H \Psi_i s_i} \right) \\
 & + N \ln |D| - \sum_{i=1}^N \left(\frac{s_i^H D s_i - s_i^H D u}{- u^H D s_i + u^H D u} \right) \\
 & + \ln |\beta_0 D| - \beta_0 \left(\frac{u^H D u - u^H D m_0}{- m_0^H D u + m_0^H D m_0} \right) \\
 & - v_0 \ln |W_0|/2 + ((v_0 + L + 1) \ln |D|)/2 \\
 & - \text{tr}(W_0^{-1} D)/2 - \ln \Gamma(a_0) \\
 & + \sum_{k=1}^D \Gamma((v_0 - k + 1)/2) + a_0 \ln b_0 \\
 & + (a_0 - 1) \ln \gamma - b_0 \gamma + \text{const.} \quad (17)
 \end{aligned}$$

a) *Update s*: In order to update s , we should calculate the expectation of (17) with respect to u , D , γ and neglect the term irrelevant to s

$$\begin{aligned}
 \langle Q(\Theta, \Omega) \rangle_{u, D, \gamma} = & LN \langle \ln \gamma \rangle \\
 & - \langle \gamma \rangle \sum_{i=1}^N \left(\frac{y_i^H y_i - y_i^H \Psi_i s_i - s_i^H \Psi_i^H y_i}{+ s_i^H \Psi_i^H \Psi_i s_i} \right) + N \langle \ln |D| \rangle \\
 & - \sum_{i=1}^N \left(\frac{s_i^H \langle D \rangle s_i - s_i^H \langle D \rangle \langle u \rangle}{- \langle u \rangle^H \langle D \rangle s_i + \langle u \rangle^H D u} \right). \quad (18)
 \end{aligned}$$

We assume that s has the following form:

$$q(s) = \prod_{i=1}^N p(s_i | \eta_i, C_i^{-1}) \quad (19)$$

where

$$\begin{aligned}
 C_i = & \langle \gamma \rangle \Psi_i^H \Psi_i + \langle D \rangle \\
 \eta_i = & C_i^{-1} (\langle \gamma \rangle \Psi_i^H s_i + \langle D \rangle \langle u \rangle). \quad (20)
 \end{aligned}$$

Hence

$$\langle s_i \rangle = \eta_i. \quad (21)$$

b) *Update u*: In order to update s , we should calculate the expectation of (17) with respect to s, γ, D and neglect the term irrelevant to u

$$\begin{aligned}
 \langle Q(\Theta, \Omega) \rangle_{s, \gamma, D} = & N \ln |\langle D \rangle| \\
 & - \sum_{i=1}^N \left(\frac{s_i^H \langle D \rangle s_i - s_i^H \langle D \rangle \langle u \rangle}{- u^H \langle D \rangle s_i + u^H \langle D \rangle u} \right) + \ln |\beta_0 \langle D \rangle| \\
 & - \beta_0 \left(\frac{u^H \langle D \rangle u - u^H \langle D \rangle m_0}{+ m_0^H \langle D \rangle m_0} \right). \quad (22)
 \end{aligned}$$

We assume that u has the following form:

$$q(u) = p(u | m, (\beta D)^{-1}) \quad (23)$$

where

$$\begin{aligned}
 \beta = & \beta_0 + N \\
 m = & \left(\beta_0 m_0 + \sum_{i=1}^N \langle s_i \rangle \right) / \beta. \quad (24)
 \end{aligned}$$

Hence

$$\langle u \rangle = m. \quad (25)$$

c) *Update D*: In order to update D , we should calculate the expectation of (17) with respect to s, γ, u and neglect the term irrelevant to D as

$$\begin{aligned}
 \langle Q(\Theta, \Omega) \rangle_{s, \gamma, u} = & N \ln |D| \\
 & - \sum_{i=1}^N \left(\frac{s_i^H \langle D \rangle s_i - s_i^H \langle D \rangle \langle u \rangle}{- \langle u \rangle^H \langle D \rangle s_i + \langle u \rangle^H D \langle u \rangle} \right) \\
 & + \ln |\beta_0 D| - \beta_0 \left(\frac{\langle u \rangle^H D \langle u \rangle - \langle u \rangle^H D m_0}{- m_0^H D \langle u \rangle + m_0^H D m_0} \right) \\
 & - (v_0 \ln |W_0|)/2 + ((v_0 + L + 1) \ln |D|)/2 \\
 & - \text{tr}(W_0^{-1} D)/2 + \sum_{k=1}^D \Gamma((v_0 - k + 1)/2) \quad (26)
 \end{aligned}$$

We assume that D has the following form:

$$q(D) = p(D | v, W) \quad (27)$$

where

$$\begin{aligned}
 v = & v_0 + 2N \\
 W = & \left[W_0^{-1} + 2 \sum_{i=1}^N C_i + 2 \sum_{i=1}^N \left(\eta_i - \sum_{i=1}^N \eta_i / N \right) \times \left(\eta_i - \sum_{i=1}^N \eta_i / N \right)^H \right. \\
 & \left. + 2 \beta_0 N \left(\frac{1}{N} \sum_{i=1}^N \eta_i - m_0 \right) \left(\frac{1}{N} \sum_{i=1}^N \eta_i - m_0 \right)^H / (\beta_0 + N) \right]^{-1}. \quad (28)
 \end{aligned}$$

Hence

$$\langle D \rangle = v W. \quad (29)$$

d) *Update γ* : In order to update γ , we should calculate the expectation of (17) with respect to $\mathbf{s}$, $\mathbf{D}$, $\mathbf{u}$ and neglect the term irrelevant to γ as

$$\begin{aligned} \langle Q(\Theta, \Omega) \rangle_{\mathbf{s}, \mathbf{D}, \mathbf{u}} &= LN \ln \gamma \\ &- \gamma \sum_{i=1}^N \left(\mathbf{y}_i^H \mathbf{y}_i - \mathbf{y}_i^H \Psi_i \mathbf{s}_i - \mathbf{s}_i^H \Psi_i^H \mathbf{y}_i \right) \\ &- \ln \Gamma(a_0) + a_0 \ln b_0 + (a_0 - 1) \ln \gamma - b_0 \gamma. \end{aligned} \quad (30)$$

We assume that γ has the following form:

$$q(\gamma) = p(\gamma | a, b) \quad (31)$$

where

$$\begin{aligned} a &= a_0 + LN \\ b &= b_0 + \sum_{i=1}^N [(\mathbf{y}_i - \Psi_i \boldsymbol{\eta}_i)^H (\mathbf{y}_i - \Psi_i \boldsymbol{\eta}_i) \\ &\quad + \text{tr}(\Psi_i^H \Psi_i \mathbf{C}_i^{-1})]. \end{aligned} \quad (32)$$

Hence

$$\langle \gamma \rangle = a/b. \quad (33)$$

e) *Termination condition*: With each iteration, we update variation distribution of all the implicit variable. If a relative change rate of the lower bound $(\text{Lb}(q(\Theta))_{\text{iter}+1} - \text{Lb}(q(\Theta))_{\text{iter}}) / \text{Lb}(q(\Theta))_{\text{iter}}$ between two successive iteration is less than 10^{-4} in the experiment, we stop the iterative process and $\boldsymbol{\eta}$ is the reconstructed time-frequency spectrum.

III. SUPERRESOLUTION TIME-FREQUENCY FEATURE EXTRACTION

A. Micromotion Characteristics Analysis

When people walk, there are various types of simultaneous movements such as the swinging motions of the arms and legs, the translation motion of the torso, etc. Since the motion of the legs and arms of a walking person is periodic, the instantaneous frequency of each periodic motion may be approximately formulated as $\sum_{i=1}^m \sum_{j=1}^n a_{ij} \sin(2\pi f_{ij}t + \varphi_{ij})$, where m and n are the number of walking person and movements existed of single person, respectively, φ_{ij} , A_{ij} , and f_{ij} denote the initial phase, the amplitude, and the frequency of the j th movement of the i th person, respectively. On the contrary, when a wheeled vehicle is moving, micro-Doppler components does not vary periodically with time and the modulation induced by the bulk component can be represented as $b \exp(j4\pi vt/\lambda)$, where v denotes the speed of the translation motion, λ denotes the radar wavelength, and b denotes the scattering coefficient of the bulk component. It is worth noting that the energy of micromotion component is much less than that of the bulk component, while the energy of a micromotion component is close to that of the torso component for the walking person.

To better understand and analyze the micro-Doppler characteristics of the walking person and moving wheeled

vehicle, we use the STFT for the joint time-frequency analysis. The time-frequency spectrum, which is denoted by $\mathbf{S} = [\mathbf{S}_{f,t}]_{f=1, t=1}^{N_F, N_T}$, is the modulus matrix of STFT of the acquisition, where f is the row index corresponding to the frequency dimension in STFT, t is the column index corresponding to the time dimension in STFT, N_F represents the length of the Fast Discrete Fourier Transformation (FDFT) for each short time signal, and N_T represents the number of samples in the time dimension in STFT.

The time-frequency spectrums without superresolution processing of a walking person and moving vehicle are given in Fig. 3(a) and (b), respectively, and the corresponding original length of FDFT is $N_F = 32$. The quadruple superresolution time-frequency spectrums are given in Fig. 3(c)–(d) and the corresponding length of FDFT is $N_F = 128$. In addition, it can be concluded from Fig. 3 that the micro-Doppler modulations induced by the swinging arms and legs will capture a wider frequency than that induced by the moving vehicle.

B. Superresolution Time-Frequency Feature

There are significant differences in the time-frequency spectra between the human and the vehicle echo, and we can extract these differences to achieve the classification. For time-frequency spectrum, higher frequency resolution means more detailed information and these details may play an important role in the subsequent classification. To make full use of the reconstructed superresolution time-frequency spectrum, we propose a novel feature extraction method in this paper to categorize radar targets into two kinds, i.e., walking person and moving vehicle.

In this section, we define superresolution time-frequency feature 1 as

$$A_1 = \text{sum}(\mathbf{G}) = \text{sum} \left([\text{abs}(G_t^1 - G_t^2)]_{t=1}^{N_T} \right) \quad (34)$$

where G_t^1 is the row index of the largest value in the t th column of $\mathbf{S}$, and G_t^2 is the row index of the second largest value in the t th column of $\mathbf{S}$. $\mathbf{G} = [\text{abs}(G_t^1 - G_t^2)]_{t=1}^{N_T}$ is the characterization of the spectral width between the maximum peak and the second peak in the instantaneous Doppler spectrum, and it reflects the dispersive degree of the main energy component in the time-frequency spectrum. For a moving vehicle, the bulk motion is the main energy component, and for the walking person, both the micromotion component and the torso component can be the main energy component. Hence, the dispersive degree of the main energy component for a moving vehicle is larger than that for the walking people, i.e., the value of A_1 for the vehicle target is larger than that for the person target. If the frequency resolution of time-frequency spectrum is low, for the two kinds of considered targets, the difference between the values of A_1 is relatively small. After the superresolution processing, since the frequency resolution is improved, for the vehicle target, G_t^1 and G_t^2 are located in the bulk component of the spectrum, and the value of $\text{abs}(G_t^1 - G_t^2)$ in each column rapidly decreases with the improvement of the frequency resolution; for the human target, G_t^1 and

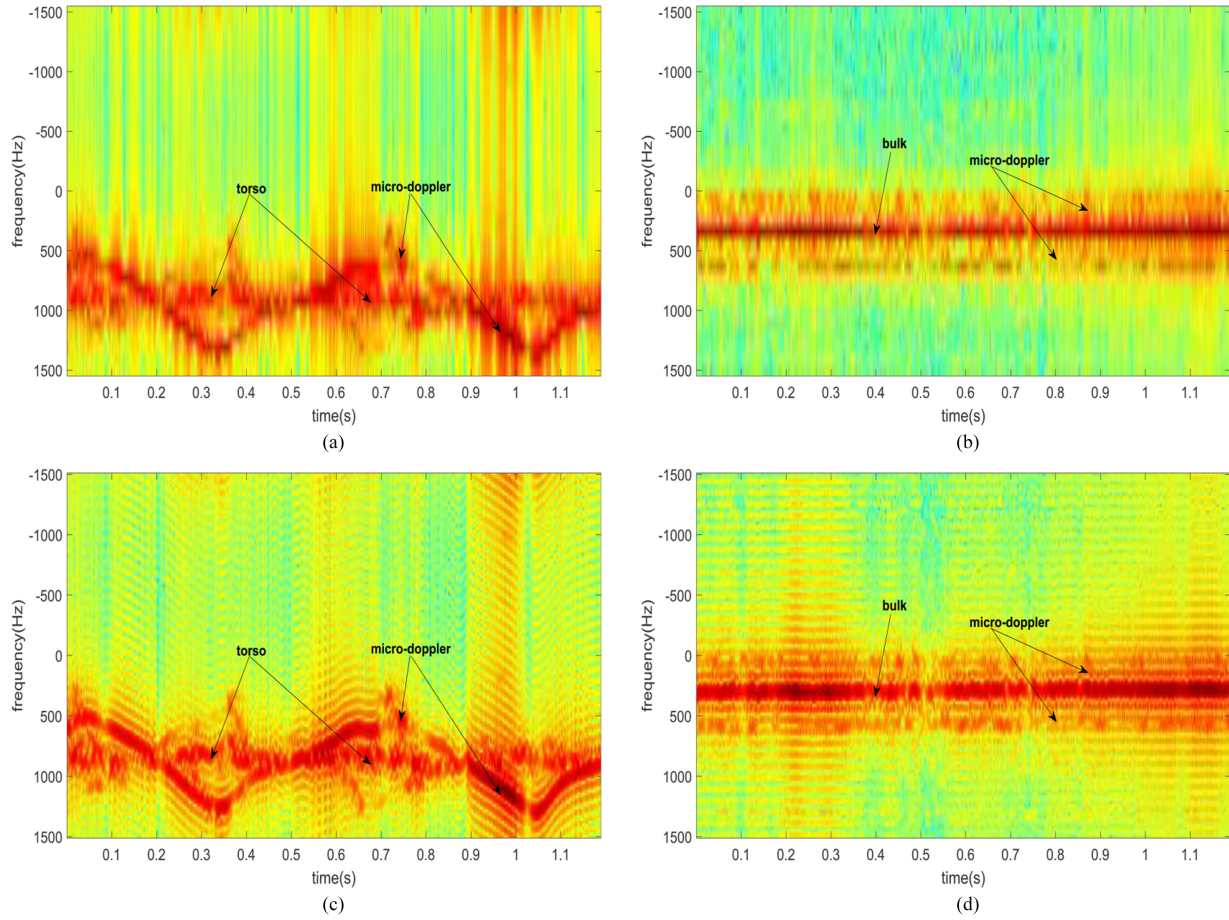


Fig. 3. Time-frequency spectrum of two kinds of targets. (a) Walking person with $N_F = 32$. (b) Moving vehicle with $N_F = 32$. (c) Walking person with $N_F = 128$. (d) Moving vehicle with $N_F = 128$.

TABLE I
Value of Feature 1 at Different Resolutions

	Original resolution	Double resolution	Tribe resolution	Quadruple resolution	Quintuple resolution
human target	3406	3347	3267	3096	3080
vehicle target	2677	2101	1789	1743	1735

G_t^2 may be located in the micromotion component and the torso component, respectively, the decline of the value of $abs(G_t^1 - G_t^2)$ in each column is relatively small. Table I gives the value of superresolution feature 1 at different resolutions. We see that the value of A_1 of the vehicle target is much less than that of the human target by the super-resolution processing, i.e., the separability between the two features from the two kinds of targets becomes better via the higher resolution.

Then, we define superresolution time-frequency feature 2 as

$$A_2 = \text{var}(\mathbf{L}) = \text{var}([L_t]_{t=1}^{N_T}) \quad (35)$$

where $L_t = \text{length}([S_{f,t} > M_t])$, $M_t = \sum_{f=1}^{N_F} S_{f,t} / N_F$ is the average of frequency modulus in the t th column of $\mathbf{S}$, and $\text{var}(\mathbf{L})$ denotes the variance of $\mathbf{L}$. The value of A_2 is determined by the fluctuation degree of $\mathbf{L}$.

We use M_t as the threshold to measure the large modulus in t th column of time-frequency spectrum, and L_t denotes the number of the large modulus in t th column of time-frequency spectrum. The bulk component captures most of the energy of the radar returns from a moving vehicle and always corresponds to the large modulus in each column of the time-frequency spectrum; however, the energy of the torso component is close to that of the micromotion component for the walking person, and the large modulus in each column of the time-frequency spectrum may correspond to both the micromotion and torso component. Besides the bandwidth of the vehicle's Doppler modulations in the time-frequency spectrum is smaller than that of walking people. Therefore, the fluctuation degree of $\mathbf{L}$ of vehicle target is less than that of walking people. After the superresolution processing, the number of the large modulus in each column of time-frequency spectrum increases, and the calculation of the large modulus is more accurate.

TABLE II
Value of Feature 2 at Different Resolutions

	Original resolution	Double resolution	Tribe resolution	Quadruple resolution	Quintuple resolution
human target	4.1469	13.0197	28.2296	49.4638	76.5839
vehicle target	1.8250	3.4689	6.7932	11.4258	15.4545

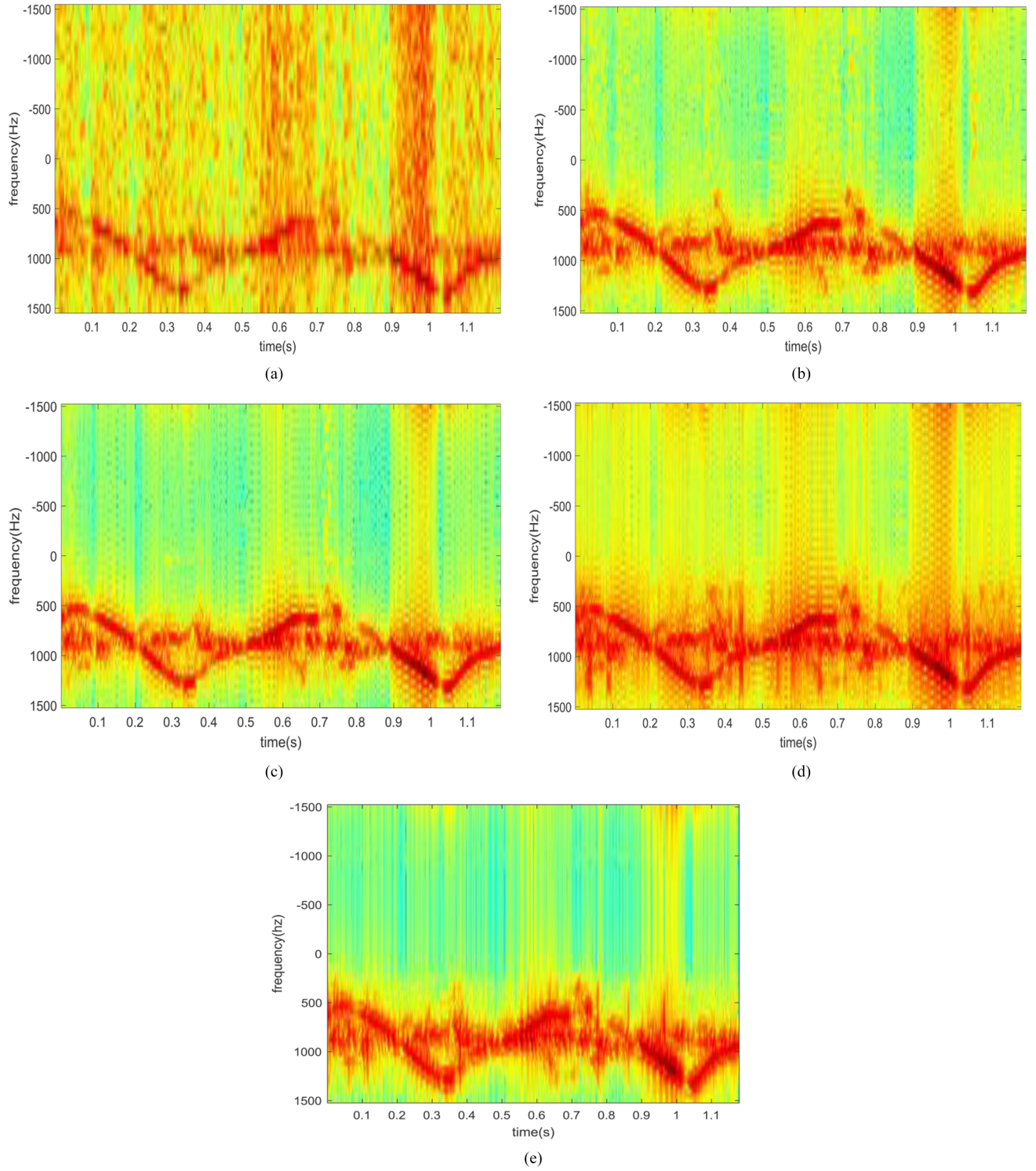


Fig. 4. Reconstruction results with 40% of data missing at random. (a) Corrupted spectrum obtained by STFT with $N_F = 64$. (b) Corresponding reconstructed spectrum of proposed method with $L = 64$. (c) Corresponding reconstructed spectrum with K-SVD. (d) Corresponding reconstructed spectrum with OMP. (e) Corresponding reconstructed spectrum with MMV.

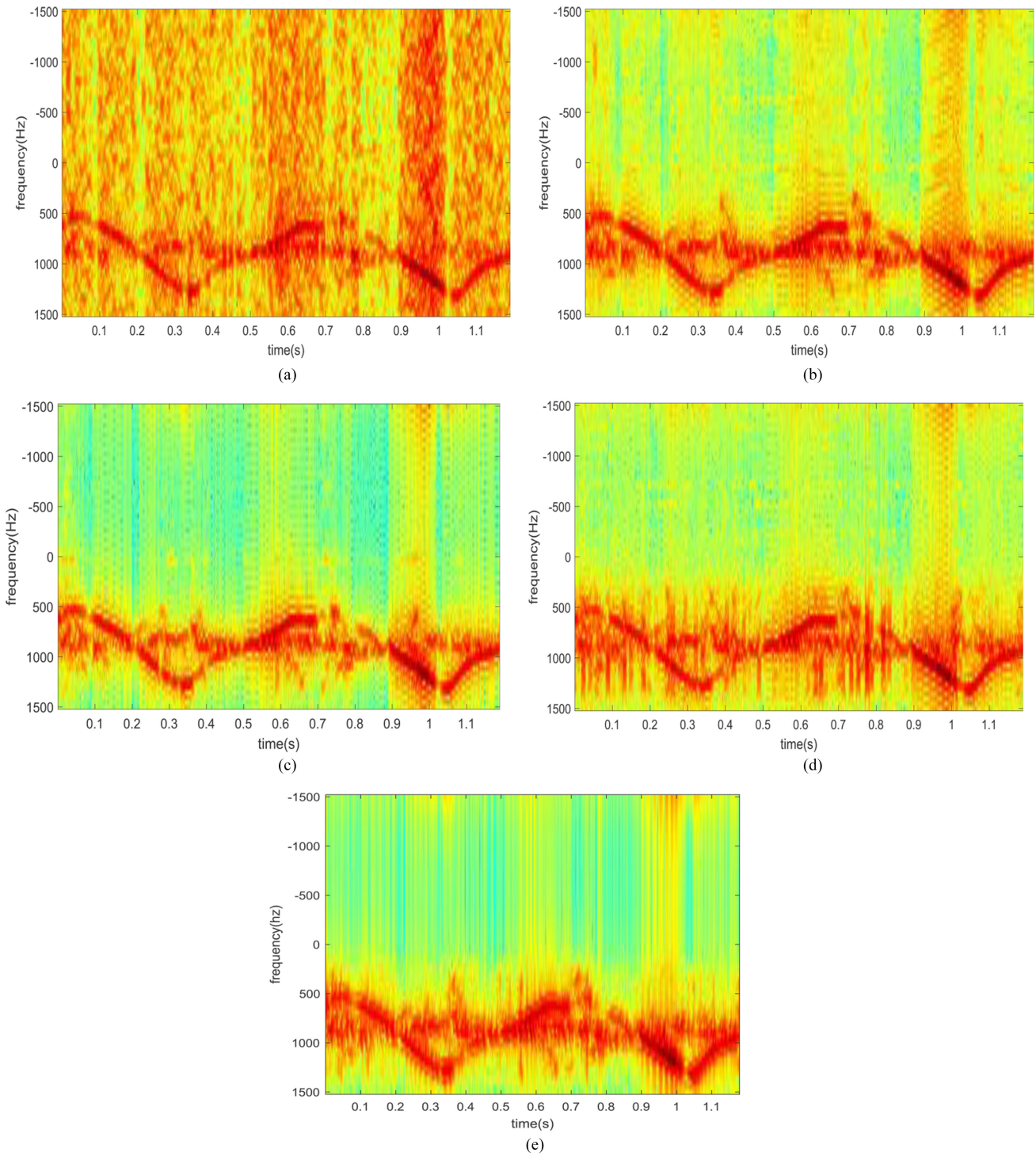


Fig. 5. Reconstruction results with 50% of data missing at random. (a) Corrupted spectrum obtained by STFT with $N_F = 64$. (b) Corresponding reconstructed spectrum of proposed method with $L = 64$. (c) Corresponding reconstructed spectrum with K-SVD. (d) Corresponding reconstructed spectrum with OMP. (e) Corresponding reconstructed spectrum with MMV.

TABLE III
Average Relative Reconstruction Errors of Different Methods With Data Missing at Random

Method	Missing rate				
	10%	20%	30%	40%	50%
Proposed method	0.0307	0.0549	0.1027	0.1235	0.1670
OMP	0.0775	0.1220	0.1731	0.2376	0.4350
K-SVD	0.0462	0.0779	0.1045	0.1415	0.1886
MMV	0.0477	0.0766	0.1084	0.1374	0.1953
Corrupted signal	0.2986	0.4349	0.5371	0.6243	0.7052

TABLE IV
Cosine Similarity of Different Methods With Data Missing at Random

Method	Missing rate				
	10%	20%	30%	40%	50%
Proposed method	0.9970	0.9957	0.9897	0.9855	0.9810
OMP	0.9900	0.9855	0.9703	0.9556	0.9435
K-SVD	0.9962	0.9941	0.9890	0.9792	0.9760
MMV	0.9963	0.9940	0.9891	0.9801	0.9760
Corrupted signal	0.9655	0.9400	0.9099	0.8717	0.8346

TABLE V
Mean Runtime for Different Algorithms With Data Missing at Random

Miss rate	Proposed method	OMP	K-SVD	MMV
10%	205.8	446.3	355.3	265.2
20%	212.4	448.5	348.1	272.4
30%	196.5	463.3	375.2	254.1
40%	198.9	433.7	359.5	260.5
50%	202.6	425.1	370.1	262.1

TABLE VI
Rényi Entropies of the Reconstructed Time-Frequency Spectra With 40% of Data Missing Random

The values of L	32	64	96	128	160
Rényi Entropy	10.1804	10.1160	9.9862	9.8734	9.8457

TABLE VII
Rényi Entropies of the Reconstructed Time-Frequency Spectra With 50% of Data Missing Random

The values of L	32	64	96	128	160
Rényi Entropy	10.1921	10.1194	10.0005	9.8792	9.8604

Because the concentration extent of time-frequency spectrum energy of the vehicle target is much higher than that of the human target, after the superresolution processing, for the vehicle target, the number of the large modulus in each column change little by the superresolution processing. For the human target, the number of the large modulus in each column has a great change. In other words, the variability degree of the large modulus in each column of human target is greater than that of the vehicle target. Table II gives the value of superresolution feature 2 at different resolutions. We can see that the value of A_2 of vehicle target is much less than that of human target by the superresolution processing, i.e., the separability between the two features from the two kinds of targets becomes better via the higher resolution.

IV. EXPERIMENT RESULTS OF RECONSTRUCTION AND CLASSIFICATION

The experimental data used in this paper are collected from the two kinds of targets, i.e., three walking people and four moving wheeled vehicles. The low-resolution radar used in the experiment is a Ka band system with its pulse repetition frequency being 3 KHz, its dwell time being 1.2 s, the pulse length being 3 μ s for each transmitted linear frequency modulated pulse, and the bandwidth being 18 MHz. There are 240 acquisitions for each kind of target.

A. Reconstruction Experiment

In this section, we utilize the measured data of walking person with two missing ways: random missing and

consistent missing. Random missing means that the diagonal elements of measurement matrix Φ are randomly selected from $\{0, 1\}$ according to data missing rate. Consistent missing means that a piece of diagonal elements of measurement matrix Φ is set to zero at a fixed interval, and the number of zero is decided by the data missing rate. We utilize the relative reconstruction error $\|\mathbf{S} - \hat{\mathbf{S}}\|_2 / \|\mathbf{S}\|_2$ to evaluate the reconstruction performance of the proposed method, where $\mathbf{S}$ is the time-frequency spectrum of the complete signal, $\hat{\mathbf{S}}$ is the reconstructed time-frequency spectrum. Besides, we also use cosine similarity to evaluate the reconstruction performance. The cosine similarity between time-frequency spectrum of the complete signal and the reconstructed time-frequency spectrum can be represented as

$$\text{similarity} = \frac{\mathbf{S} \cdot \hat{\mathbf{S}}}{\|\mathbf{S}\| \cdot \|\hat{\mathbf{S}}\|}. \quad (36)$$

The resulting similarity ranges from -1 meaning exactly opposite, to 1 meaning exactly the same, with 0 indicating orthogonality or decorrelation, while in-between values indicate intermediate similarity or dissimilarity.

In our experiment, we consider the reconstruction of double superresolution time-frequency spectrum, i.e., $L = 64$ and $M = 32$. As a contrast, we also use K -SVD algorithm [24], which is a dictionary learning method based on K -SVD, the OMP algorithm [21], and the MMV algorithm [22] to realize reconstruction. We use the publicly available code for these algorithms indicated above and use the given parameter settings. It is worth noting that OMP algorithm and K -SVD algorithm realize the

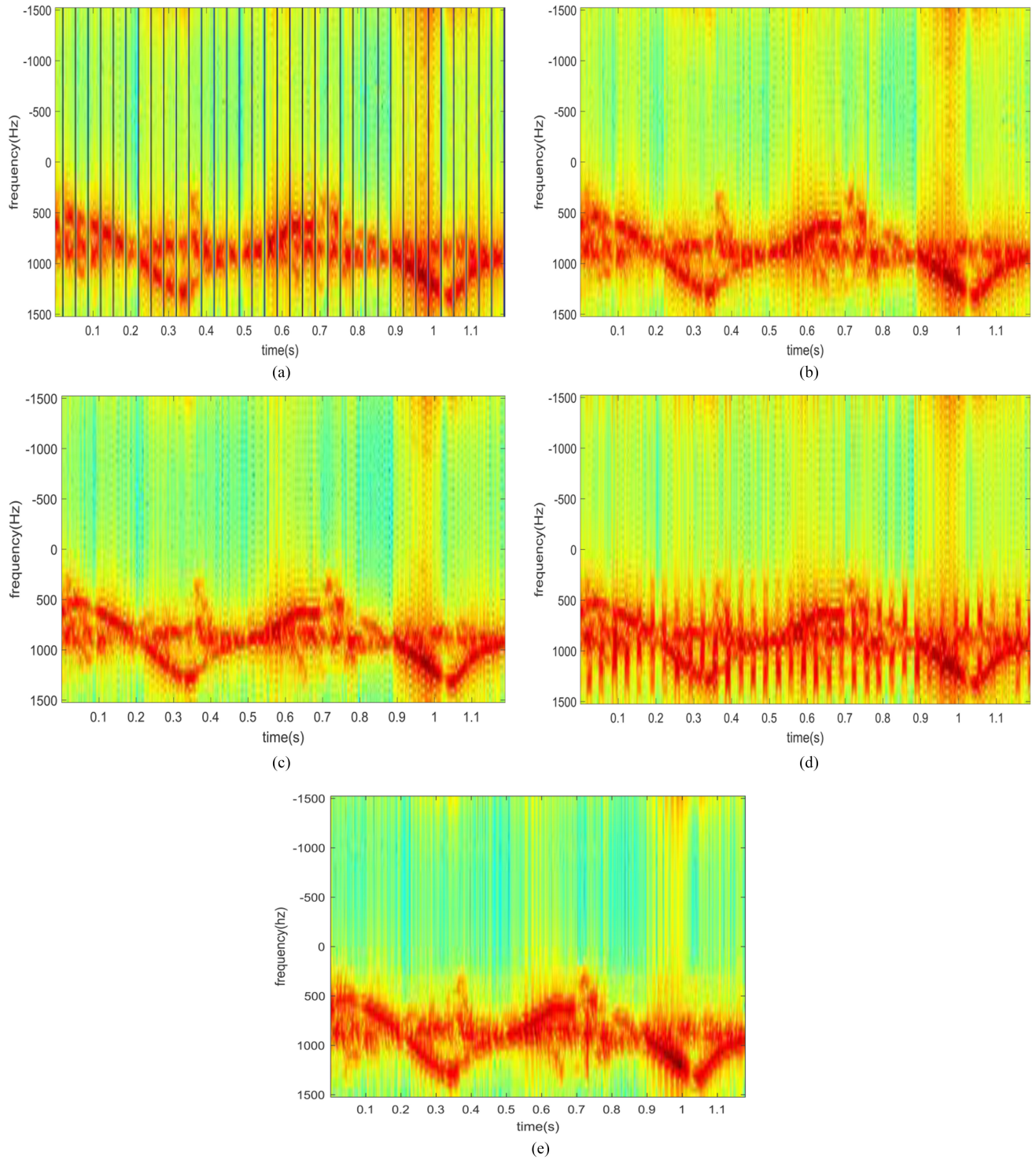


Fig. 6. Reconstruction results with 40% of data missing consistently. (a) Corrupted spectrum obtained by STFT with $N_F = 64$. (b) Corresponding reconstructed spectrum of proposed method with $L = 64$. (c) Corresponding reconstructed spectrum with K -SVD. (d) Corresponding reconstructed spectrum with OMP. (e) Corresponding reconstructed spectrum with MMV.

reconstruction of time-domain signal, and the corresponding time-frequency spectra are obtained by STFT; meanwhile, the corrupted spectra are also obtained by STFT. Here, we think that one method can achieve good reconstruction result when the relative reconstruction error is no more than 0.2.

1) *Random Missing Experiment:* Fig. 4 shows the corrupted time-frequency spectrum of walking person with $N_F = 64$ and the reconstructed time-frequency spectra when data missing rate is 40%. Fig. 5 shows the experimental results when the data missing rate is 50%. The spectra given in Figs. 4 and 5 are the results in one experiment with

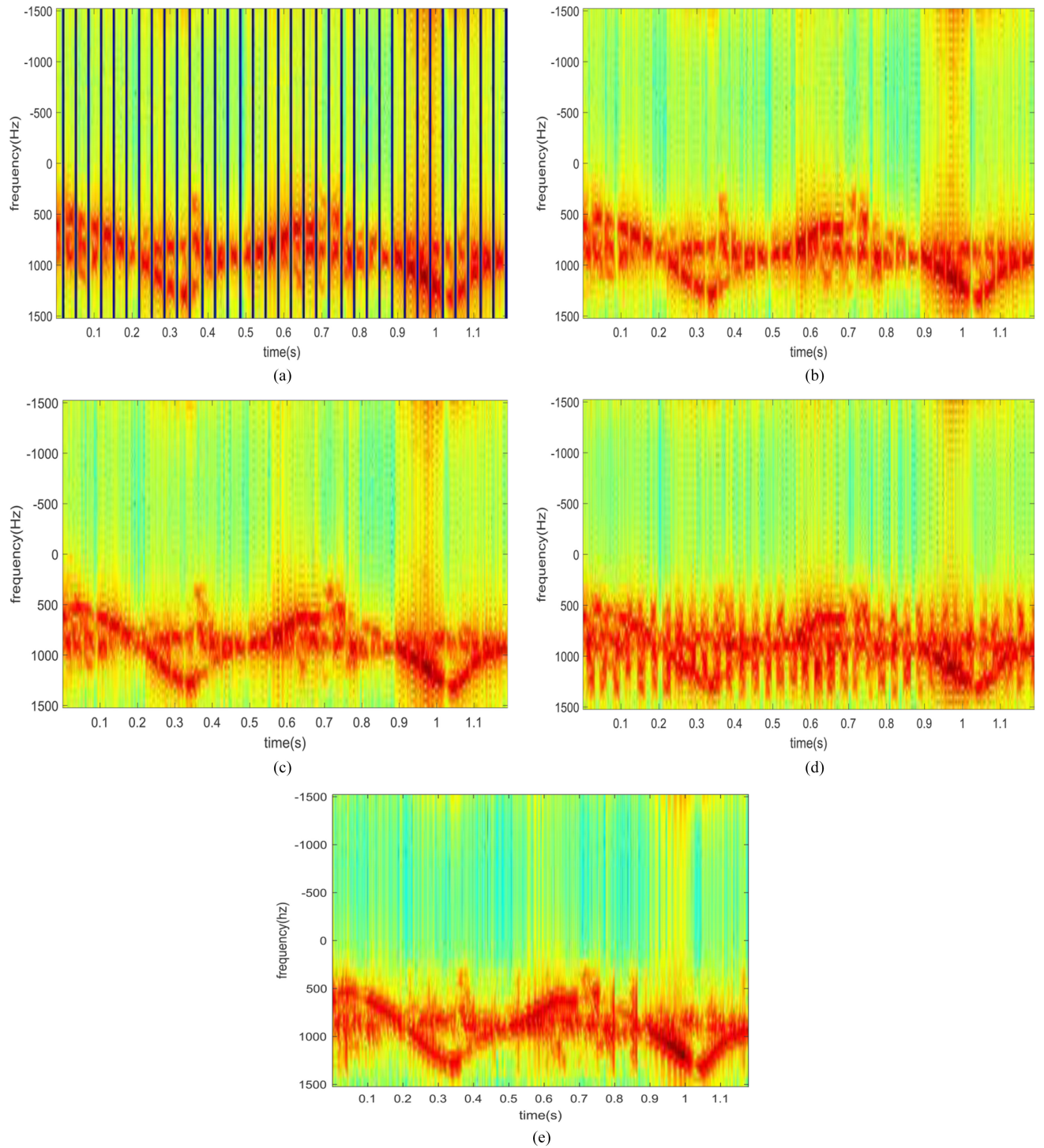


Fig. 7. Reconstruction results with 50% of data missing consistently. (a) Corrupted spectrum obtained by STFT with $N_f = 64$. (b) Corresponding reconstructed spectrum of proposed method with $L = 64$. (c) Corresponding reconstructed spectrum with K -SVD. (d) Corresponding reconstructed spectrum with OMP. (e) Corresponding reconstructed spectrum with MMV.

data missing at random. Tables III and IV presents the relative reconstruction errors and cosine similarity of walking person under different data missing rates, respectively, each of which is averaged over 40 samples.

As can be seen from Figs. 4(b) and 5(b), the proposed method can effectively reconstruct the time-frequency

spectrum of corrupted signal. It is evident from Tables III and IV that the reconstruction performance of the proposed method is better than the OMP method with data missing at random. When the missing rate is more than 30%, it is hard to reconstruct spectrum for the OMP method while the proposed method can still perform well. Although the

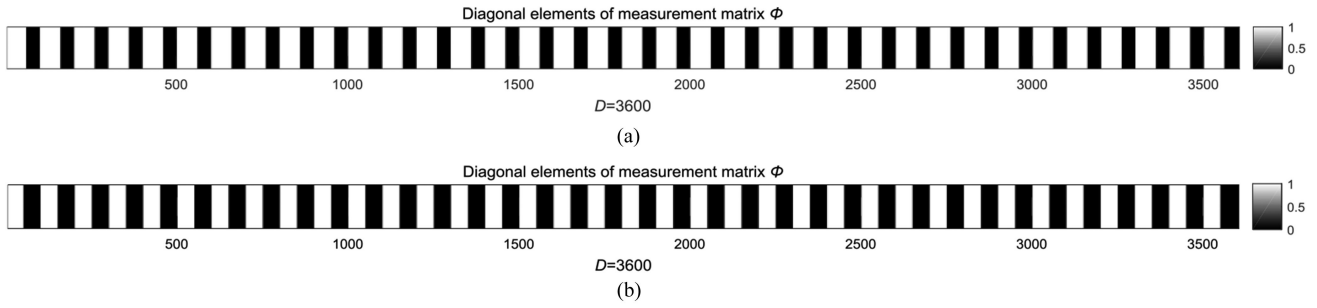


Fig. 8. Two examples of the diagonal elements of measurement matrix Φ with data missing consistently. (a) Setting 40 consecutive diagonal elements to zero by every hundred-element intervals. (b) Setting 50 consecutive diagonal elements to zero by every hundred-element intervals.

TABLE VIII
Average Relative Reconstruction Errors of Different Methods With Data Missing Consistently

Method	Missing rate				
	10%	20%	30%	40%	50%
Proposed method	0.0801	0.1353	0.2339	0.2859	0.3818
OMP	0.2844	0.3943	0.5561	0.6854	0.7520
K-SVD	0.1336	0.1942	0.2397	0.3159	0.3958
MMV	0.1369	0.2033	0.2511	0.3213	0.3970
Corrupted signal	0.3132	0.4828	0.5713	0.6877	0.7421

TABLE IX
Cosine Similarity of Different Methods With Data Missing Consistently

Method	Missing rate				
	10%	20%	30%	40%	50%
Proposed method	0.9949	0.9871	0.9725	0.9700	0.9394
OMP	0.9722	0.9512	0.9100	0.8554	0.7822
K-SVD	0.9892	0.9760	0.9714	0.9462	0.9350
MMV	0.9870	0.9729	0.9700	0.9446	0.9325
Corrupted signal	0.9413	0.9280	0.9020	0.8553	0.8065

reconstruction performance of K -SVD and MMV is close to our algorithm, we can still see an improvement using the proposed method.

Table V shows the mean runtime for different algorithms with data missing at random. Our experiments are executed on Intel Core CPU i7-6700 @ 3.40 GHz 3.41 GHz. The RAM memory is 16 GB. We note that our algorithm has an obvious advantage in computing speed.

The superresolution spectrum means the more energy-concentrated time-frequency spectrum. The Rényi entropy is a useful indicator to evaluate the energy concentration of a time-frequency spectrum. The Rényi entropy with order α is defined as

$$H_{\alpha}(X) = \frac{\alpha}{1-\alpha} \sum_{i=1}^n p_i^{\alpha} \quad (37)$$

where X is a discrete random variable, and p_i is corresponding probabilities. Here, we utilize the Rényi entropy with $\alpha = 2$ to evaluate the performance of a superresolution technique, and a lower Rényi entropy value means a more energy-concentrated time-frequency spectrum. The Rényi entropies of the time-frequency spectra reconstructed by our method are given in Tables VI and VII with different data missing rates. It can be seen that reconstructed time-frequency spectrum with $L = 160$ has the lowest Rényi

entropy, which denotes that it is the highest-resolution time-frequency spectrum.

2) *Consistent Missing Experiment:* Fig. 6 shows the reconstruction results with 40% of data missing consistently. Fig. 7 shows the experimental results with 50% of data missing consistently. We give two examples of the diagonal elements of measurement matrix $\Phi \in \mathbb{R}^{D \times D}$ ($D = 3600$) with data missing consistently: setting 40 consecutive diagonal elements to zero by every hundred-element intervals and setting 50 consecutive diagonal elements to zero by every hundred-element intervals, whose sketch maps are shown in Fig. 8. For the two examples, the corresponding missing rates are 40% and 50%, respectively. Tables VIII and IX give the relative reconstruction errors and cosine similarity of a walking person under different data missing rates, respectively, each of which is averaged over 40 samples. It is obvious that the OMP method is not almost able to realize reconstruction. We can see that the proposed method can still realize good reconstruction when the missing rate is no more than 20% and outperforms other methods.

B. Classification Experiment

In this section, we utilize the measured data of two kinds of targets with random missing to evaluate the

TABLE X
Classification Accuracies of the Different Features With Different Missing Rates and Different Values of L (Different Frequency Resolutions for Reconstructing the Time-Frequency Spectra)

Missing rates	Method	The values of L				
		32	64	96	128	160
10%	super-resolution feature	93.75%	95.83%	97.50%	97.50%	97.92%
	time-frequency feature	94.17%	94.17%	94.17%	95.00%	95.00%
	EMD feature	86.67%	87.08%	86.67%	86.67%	87.08%
20%	super-resolution feature	92.08%	95.00%	97.08%	97.22%	97.92%
	time-frequency feature	93.50%	93.50%	94.10%	93.75%	94.10%
	EMD feature	85.83%	85.83%	85.83%	85.83%	85.83%
30%	super-resolution feature	89.13%	94.17%	95.00%	96.67%	97.50%
	time-frequency feature	90.00%	90.83%	91.67%	91.67%	91.67%
	EMD feature	85.00%	84.58%	84.58%	84.58%	85.00%
40%	super-resolution feature	88.75%	92.92%	94.17%	96.25%	96.67%
	time-frequency feature	88.33%	89.58%	89.58%	89.58%	89.58%
	EMD feature	83.33%	82.92%	93.75%	82.92%	82.92%
50%	super-resolution feature	86.25%	91.67%	92.08%	93.33%	93.75%
	time-frequency feature	87.50%	87.92%	87.50%	87.50%	87.92%
	EMD feature	80.83%	80.83%	80.83%	80.42%	80.83%

classification performance of the proposed superresolution feature. We extract feature from the reconstructed time-frequency spectrum using proposed method at different resolutions, and the SVM is used to classify the two kinds of targets. For each kind of targets, we use half of the data as the training data and the rest as the test data. For comparison, the time-frequency feature proposed in [30] and EMD feature proposed in [31] are also used to perform classification. Table X gives the classification accuracies via different features.

It can be clearly seen from Table X that the classification accuracies of the three kinds of features gradually decrease with the increase of the missing rate. When $L = 32$, the average classification accuracy of the superresolution feature is close to that of a time-frequency feature and is higher than an EMD feature. By raising the value of L to 160, the classification performance of the superresolution feature has been improved significantly while the classification performance of the time-frequency feature has no significant improvements and the classification performance of EMD feature changes little. We can conclude that the classification accuracy of the proposed feature maintains higher than 95% when the missing rate is no more than 30% and $L \geq 96$, and when even the missing rate is up to 50%, the classification accuracy of proposed feature is still up to 93.75% by the superresolution processing with $L = 160$ while the classification accuracy of the time-frequency feature without superresolution is 87.50%. Therefore, the proposed superresolution feature with superresolution reconstruction outperforms time-frequency feature with su-

perresolution reconstruction for the same corrupted data, and the classification performance of the superresolution feature can be improved greatly by the superresolution processing.

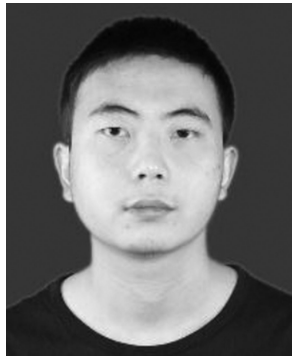
V. CONCLUSION

In this paper, we propose a novel signal reconstruction model based on the SCS frame to overcome the signal degeneracy problem. Then, the superresolution time-frequency feature is extracted to categorize ground moving targets into two kinds, i.e., walking person and a moving wheeled vehicle. It has been demonstrated by experiments on measured data that the proposed model has good reconstruction performance and the proposed time-frequency feature provides good potential for ground moving target classification.

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His research interests include radar signal processing and radar automatic target recognition.



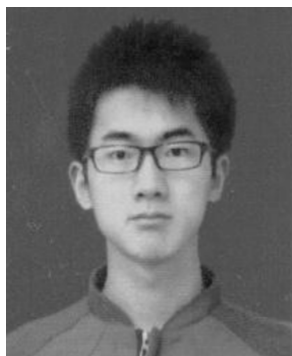
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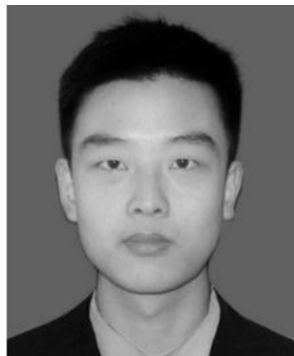
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